# 大学物理

电子教案

清华大学物理系基础物理教研组

# 第三章 动量与角动量

§ 3.1 冲量与动量定理
§ 3.2 质点系的动量
§ 3.3 动量守恒定律
§ 3.4 火箭飞行原理
§ 3.5 质心
§ 3.6 质心运动定理
§ 3.7 两体问题
§ 3.8 质点的角动量
§ 3.9 角动量守恒定律

# § 3.1 冲量（impulse）与动量（momentum）定理

动量 $\boxed{\vec{p} \equiv m\vec{\upsilon}}$

动量定理 $\Delta\vec{I} = \vec{F}\Delta t = \Delta\vec{p}$

力的时间积累，即冲量

有限时间内 (变化的力)， **initial -----> final**

$$\vec{I} = \sum_{j=1}^{n} \vec{F}(t_j)\Delta t_j = \int_{t_i}^{t_f} \vec{F}(t)dt = \int_{t_i}^{t_f} \frac{d\vec{p}}{dt} dt = \vec{p}_f - \vec{p}_i$$

平均冲力

$$\left\langle \vec{F} \right\rangle = \frac{\sum_{j=1}^{n} \vec{F}(t_j)\Delta t_j}{t_f - t_i} = \frac{\vec{p}_f - \vec{p}_i}{t_f - t_i}$$

例：一篮球质量**0.58kg**，从**2.0m**高度下落，到达地面后，以同样速率反弹，接触时间仅**0.019s**，

求：对地平均冲力？

解：篮球到达地面的速率

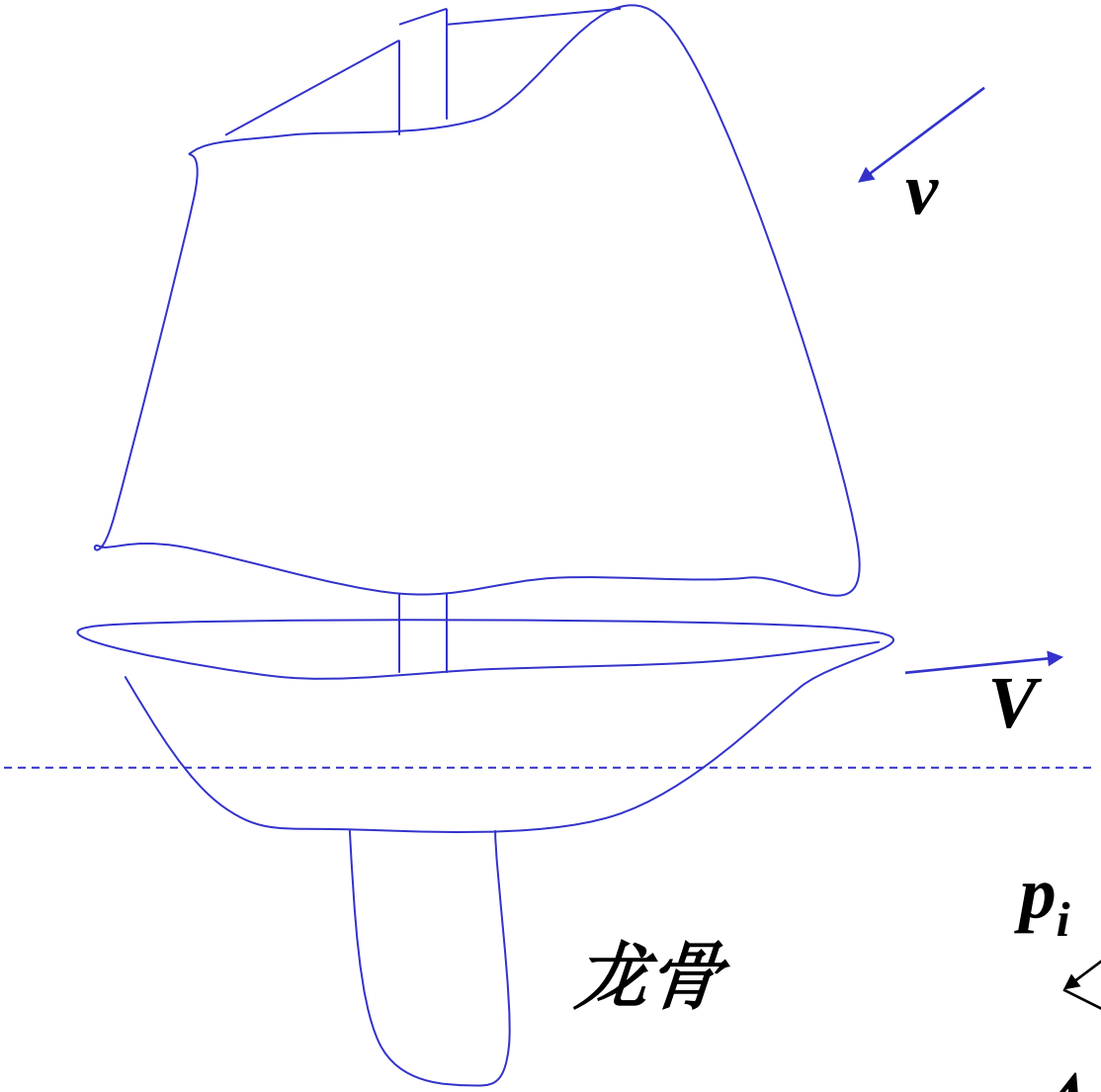
$$v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 2} = 6.3\text{（m/s）}$$

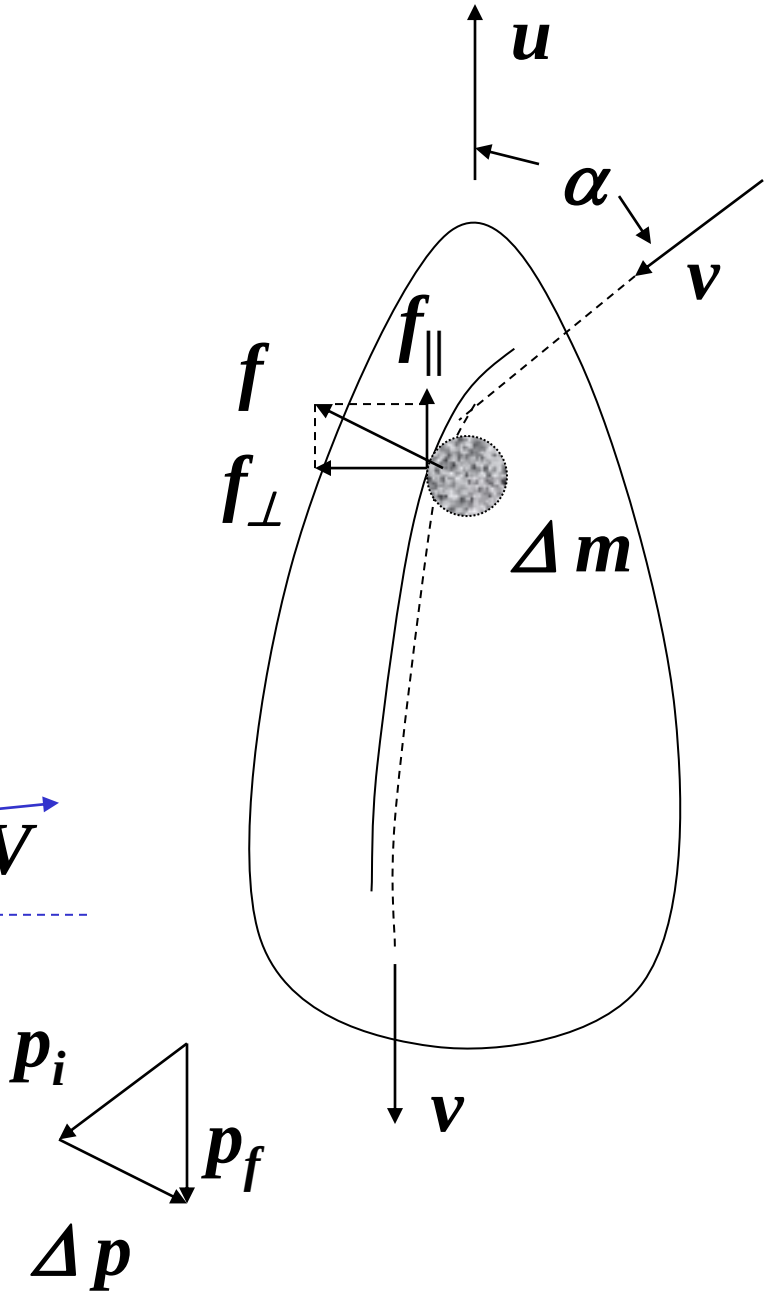
$$\bar{F}\text{或}\left\langle F \right\rangle = \frac{2mv}{\Delta t} = \frac{2 \times 0.58 \times 6.3}{0.019} = 3.8 \times 10^2\text{（N）}$$

哥伦比亚号航天飞机

失事原因：发射时泡沫高速撞击，损毁表面，返回时损毁表面在高温下瓦解。

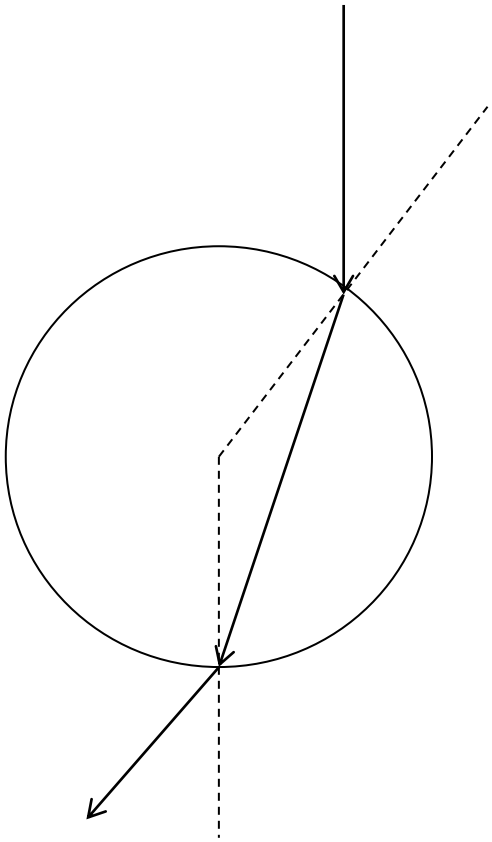
# 例：逆风行舟



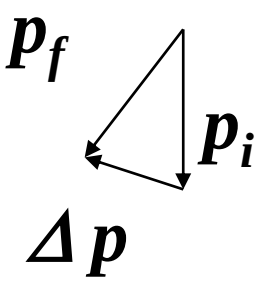
# 光镊

小物体透明（反射相对弱很多）

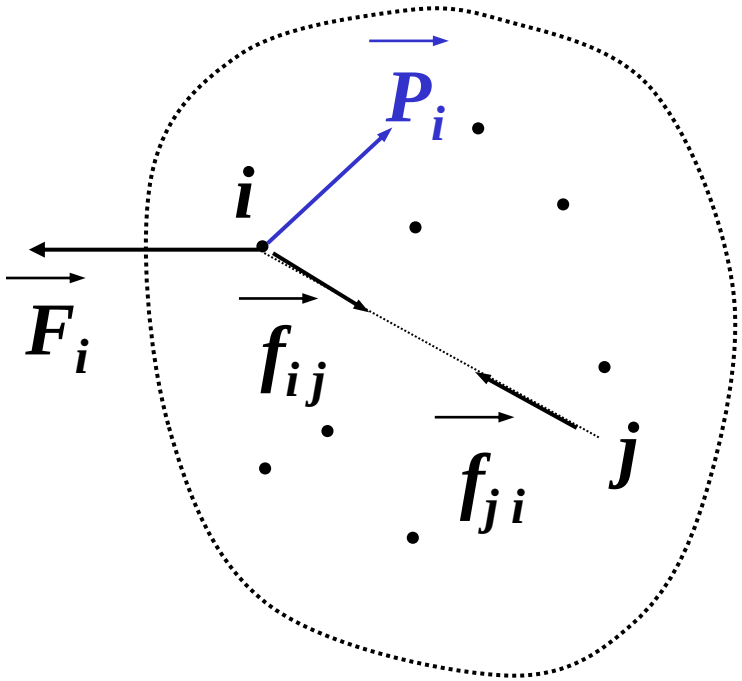


## 光子动量

$$p = \frac{h}{\lambda} = \frac{h\nu}{c}$$



# § 3.2 质点系的动量




共有*N*个粒子，外力用 ***F***，内力（即粒子之间的相互作用）用 ***f***，则第 ***i*** 粒子的运动方程

$$\vec{F}_i + \sum_{i \neq j} \vec{f}_{ij} = \frac{d\vec{p}_i}{dt}$$

对所有粒子求和

$$\sum_{i=1}^{N} \vec{F}_i + \sum_{i=1}^{N} \sum_{i \neq j} \vec{f}_{ij} = \frac{\sum_{i=1}^{N} d\vec{p}_i}{dt} = \frac{d \sum_{i=1}^{N} \vec{p}_i}{dt}$$

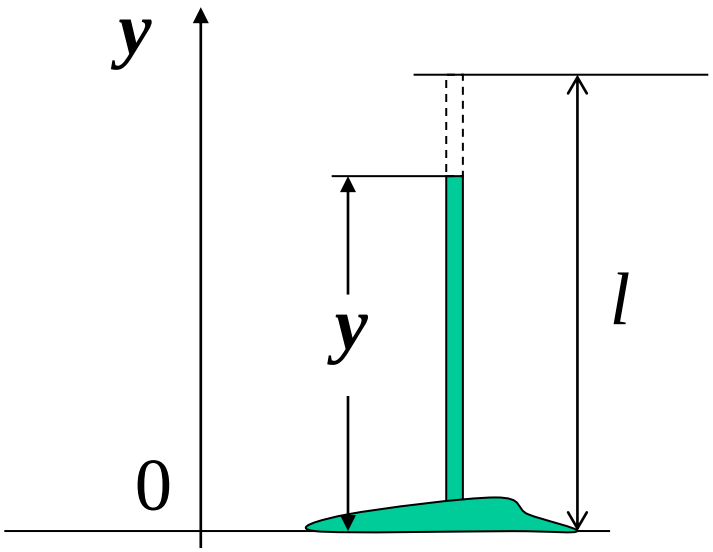
$$\vec{F} = \sum_{i=1}^{N} \vec{F}_i \qquad \vec{P} = \sum_{i=1}^{N} \vec{p}_i$$

$$\sum_{i=1}^{N} \sum_{i \neq j} \vec{f}_{ij} = \sum_{i=1}^{N} \sum_{i > j} (\vec{f}_{ij} + \vec{f}_{ji})$$

牛顿第三定律 $\vec{f}_{ij} + \vec{f}_{ji} = 0$

质点系 $\boxed{\vec{F} = \dfrac{d\vec{P}}{dt}}$ $\qquad d\vec{I} = \vec{F}dt = d\vec{P}$

例：一柔软绳长 $l$，线密度 ρ，一端着地开始自由下落，下落的任意时刻，给地面的压力为多少？




设压力为 $N$

解：在竖直向上方向建坐标，地面为原点（如图）。

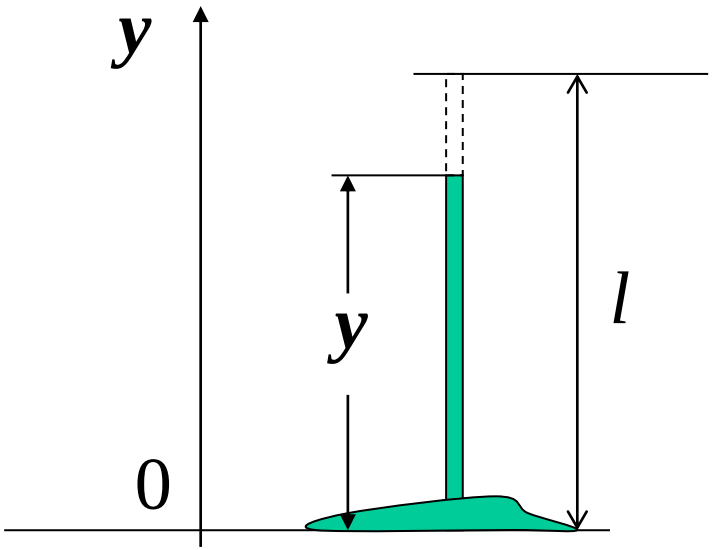
$$N-\rho gl=\frac{dp}{dt}=\dot{p}$$

$$p=\rho yv \quad -> \quad \frac{dp}{dt}=\rho\frac{d(yv)}{dt}$$

$$N=\rho gl+\rho\frac{d(yv)}{dt}$$

$$N = \rho g l + \rho \frac{d(yv)}{dt}$$

$$v = \frac{dy}{dt} \qquad -g = \frac{dv}{dt}$$




$$\frac{d(yv)}{dt} = -yg + v^2$$

$$v = -gt \qquad y = l - \frac{1}{2} g t^2$$

$$l - y = \frac{v^2}{2g} \quad \rightarrow \quad v^2 = 2g(l - y)$$

$$N = 3\rho g(l - y)$$

上层垮塌，承重陡增3倍，压垮整个建筑？

# §3.3 动量守恒定律

质点系所受合外力为零，总动量不随时间改变，即

$$\vec{P} = \sum_{i=1}^{N} \vec{p}_i = \text{常矢量}$$

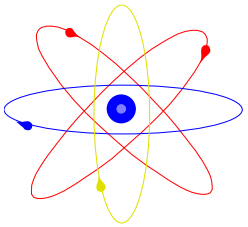
1. 合外力为零，或外力与内力相比小很多；

2. 合外力沿某一方向为零；　$\sum_i p_{i\alpha} = const.$

3. 适用于惯性系；

4. 比牛顿定律更普遍的最基本的定律。

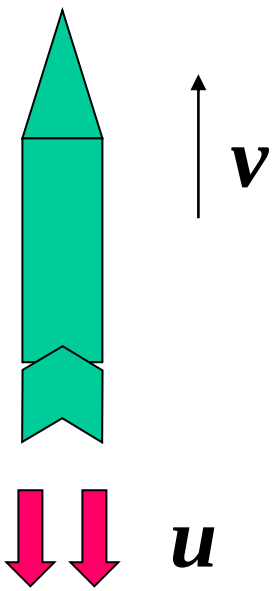
（空间的平移对称性相关）



# § 3.4 火箭飞行原理

***u*** 燃料相对火箭速度

$$-Mg = \frac{dp}{dt}$$




$$p(t) = Mv$$

$$p(t+dt) = (M - dm)(v + dv) + dm(v - u)$$

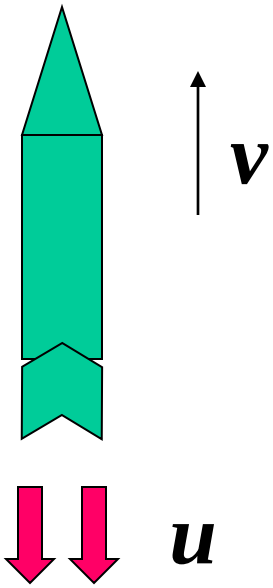
$$\therefore dp = Mdv - udm$$

$$u\frac{dm}{dt} - Mg = M\frac{dv}{dt}$$

$$or \qquad u\dot{m} - Mg = M\dot{v}$$

$$-dM = dm$$

$$-u\frac{dM}{dt} - Mg = M\frac{d\upsilon}{dt}$$



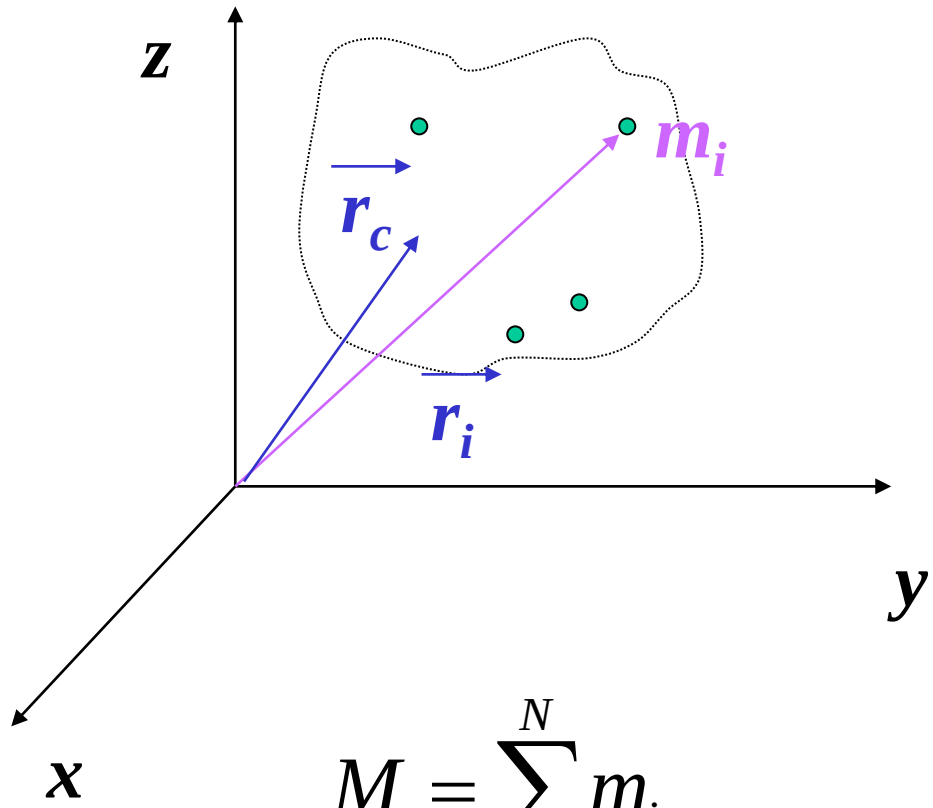

$$-u\frac{dM}{M} - gdt = d\upsilon$$

$$\int_{\upsilon_i}^{\upsilon_f} d\upsilon = -u\int_{M_i}^{M_f}\frac{dM}{M} - g\int_{t_i}^{t_f} dt$$

$$\upsilon_f - \upsilon_i = u\ln\frac{M_i}{M_f} - g(t_f - t_i)$$

# § 3.5 质心

*N*个粒子系统，可定义质量中心




$$\vec{r}_c = \frac{\sum_{i=1}^{N} m_i \vec{r}_i}{\sum_{i=1}^{N} m_i} = \frac{\sum_{i=1}^{N} m_i \vec{r}_i}{M}$$

$$x_c = \frac{\sum_{i=1}^{N} m_i x_i}{M}$$

$$M = \sum_{i=1}^{N} m_i$$

同理对 *y* 和 *z* 分量

对连续分布的物质，可以将其分为*N*个小质元

$$x_c = \frac{\sum_{i=1}^{N} x_i \Delta m_i}{M} = \frac{\int x dm}{M}$$

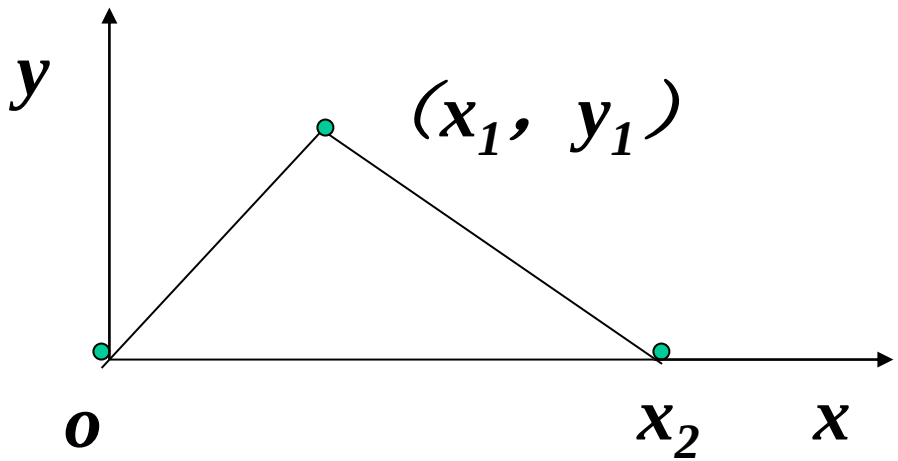
$$M = \int dm$$

同理对 *y* 和 *z* 分量

这样定义的质心有很多特性

# 质量均匀分布时，质心是几何中心

例：任意三角形的每个顶点有一质量$m$，求质心。




$$x_c = \frac{mx_1 + mx_2}{3m} = \frac{x_1 + x_2}{3}$$

$$y_c = \frac{my_1}{3m} = \frac{y_1}{3}$$

对均匀质量分布的三角形板或圆盘都可以验算

$$\vec{r}_c = \frac{\sum_{i=1}^{N} m_i \vec{r}_i}{M}$$

求导

$$\dot{\vec{r}}_c = \frac{\sum_{i=1}^{N} m_i \dot{\vec{r}}_i}{M}$$

质心速度

$$\vec{v}_c = \frac{\sum_{i=1}^{N} m_i \vec{v}_i}{M}$$

$$\boxed{\sum_{i}^{N} \vec{p}_i = M\vec{v}_c}$$

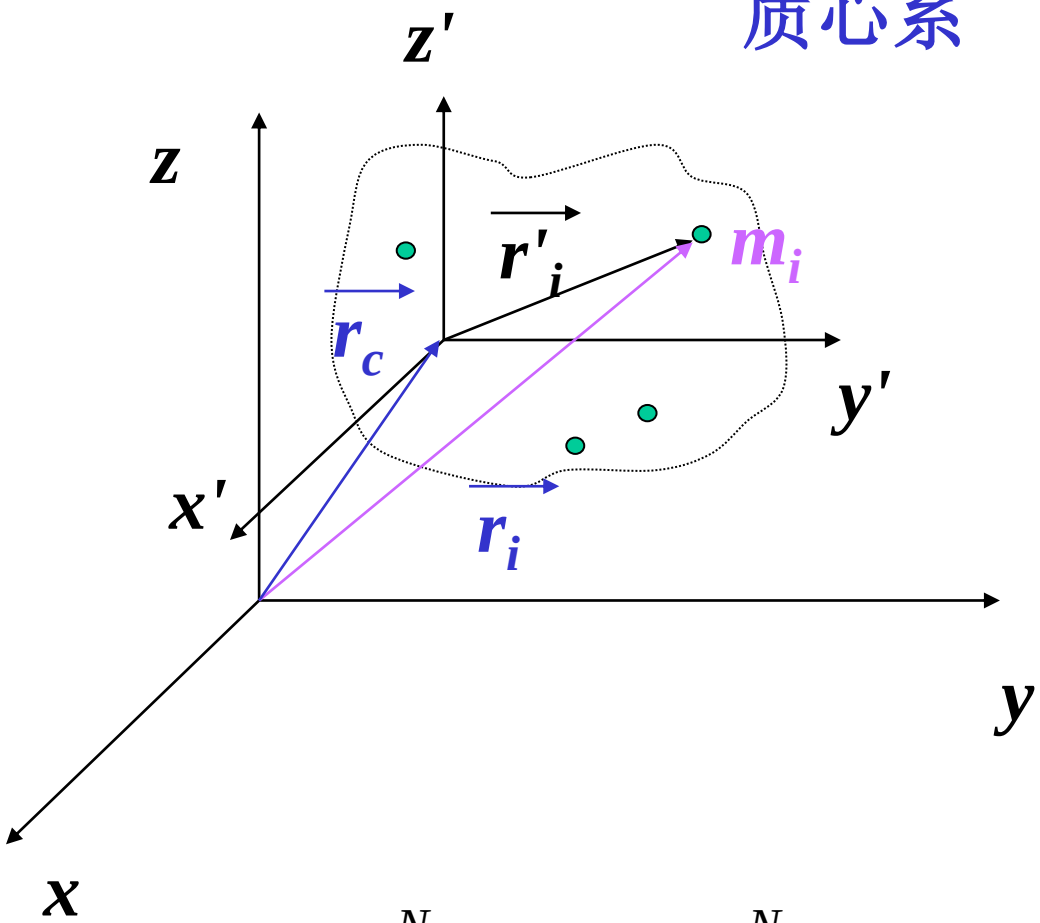
质点系的总动量是质心动量

再求导

质心加速度

$$\vec{a}_c = \frac{\sum_{i=1}^{N} m_i \vec{a}_i}{M}$$

# 质心系




质心参考系

质心坐标系

原点选在质心

$$\vec{r}_i' = \vec{r}_i - \vec{r}_c$$

$$\sum_{i=1}^{N} m_i \vec{r}_i' = \sum_{i=1}^{N} m_i (\vec{r}_i - \vec{r}_c) = 0$$

$$\boxed{\sum_{i=1}^{N} m_i \vec{r}_i{}' = 0}$$

求导 $\sum_{i=1}^{N} m_i \vec{v}_i{}' = 0$

质心系可能不是惯性系，但质心系特殊，动量守恒定律适用，而且，总动量 = 0。

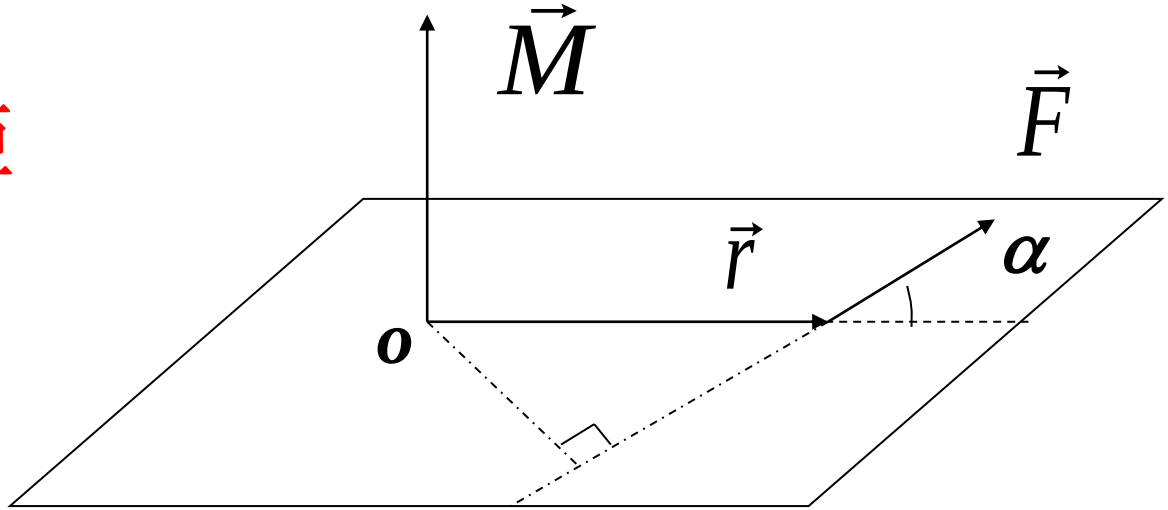
$$\vec{r}_i{}' = \vec{r}_i - \vec{r}_c$$

质心系中的速度 $\vec{v}_i{}' = \vec{v}_i - \vec{v}_c$

再求导 $\sum_{i=1}^{N} m_i \vec{a}_i{}' = 0$ $\vec{a}_i{}' = \vec{a}_i - \vec{a}_c$

重心

力矩



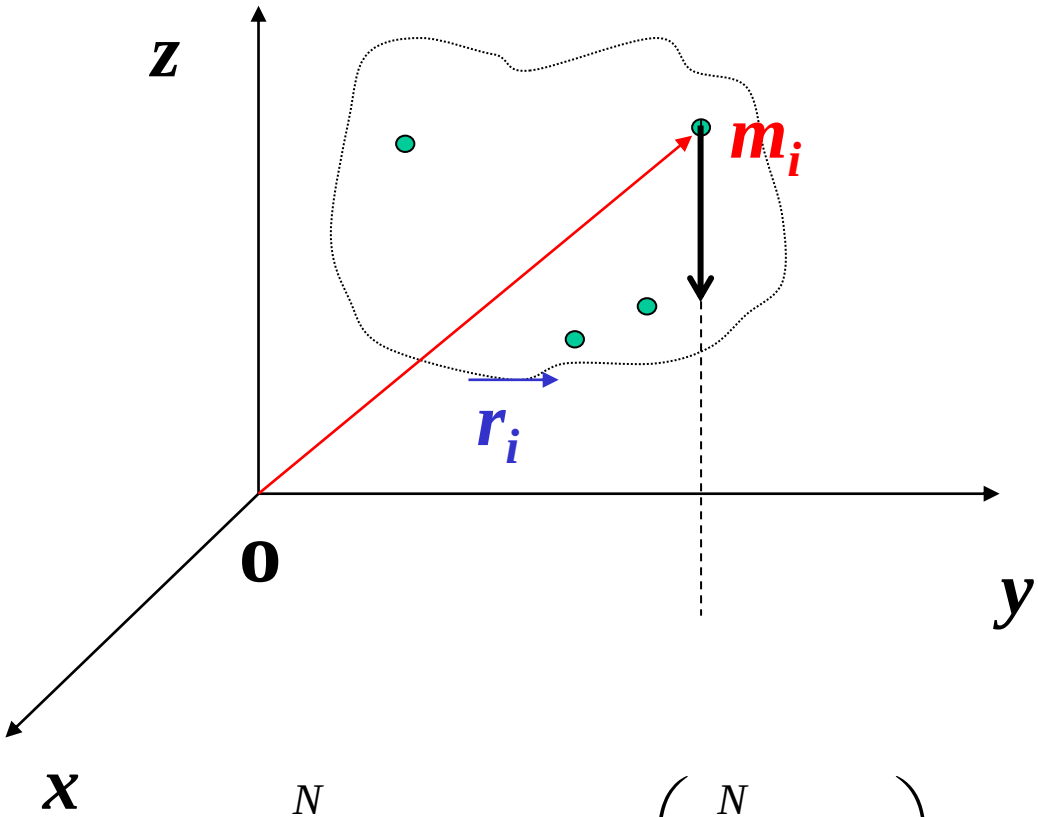

$$rF\sin\alpha$$

$$\boxed{\vec{M} = \vec{r} \times \vec{F}}$$

方向用右手螺旋法规定

力矩：赝矢量

# 地面上的重心




相对原点的重力矩

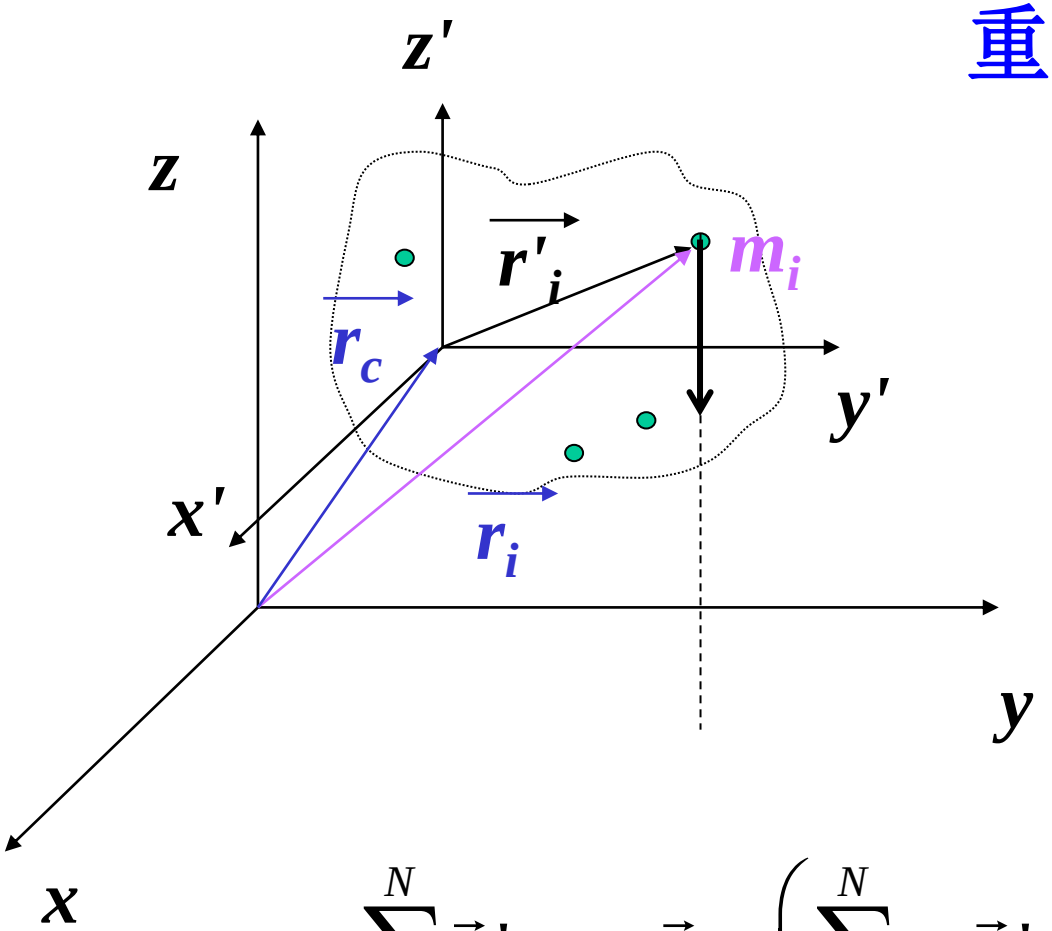
$$r_i m_i g \sin\theta$$

$$\vec{r}_i \times m_i \vec{g}$$

$$\sum_{i=1}^{N} \vec{r}_i \times m_i \vec{g} = \left( \sum_{i=1}^{N} m_i \vec{r}_i \right) \times \vec{g} = M\vec{r}_c \times \vec{g} = \vec{r}_c \times M\vec{g}$$

相当于所有质量集中于质心产生的力矩

# 重心



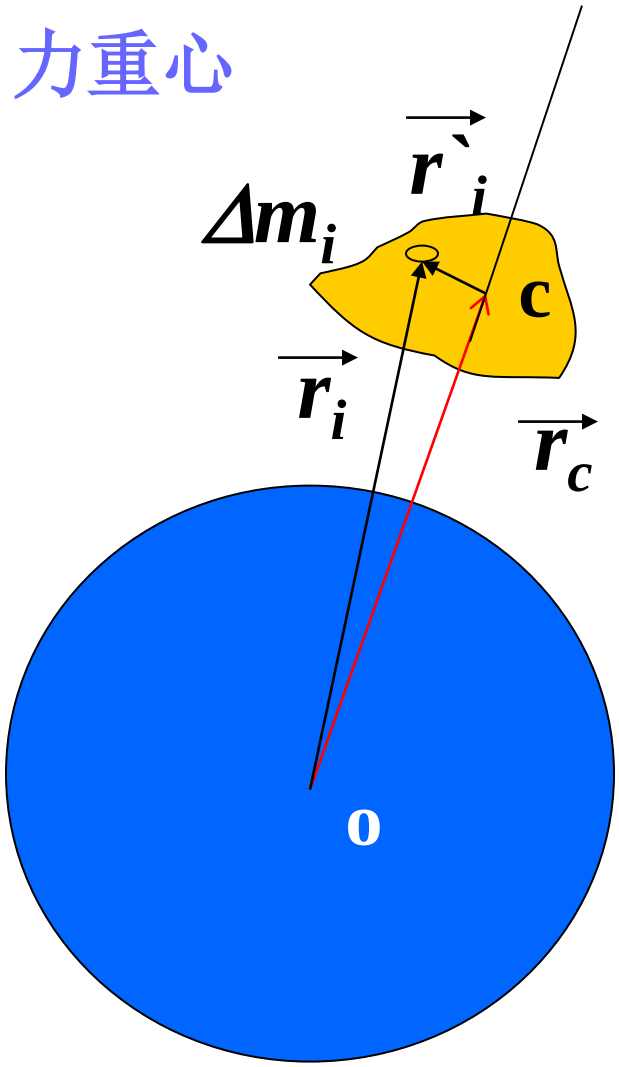

相对质心的力矩

$$\vec{r}_i' \times m_i \vec{g}$$

$$\sum_{i=1}^{N} \vec{r}'_i \times m_i \vec{g} = \left( \sum_{i=1}^{N} m_i \vec{r}'_i \right) \times \vec{g} \quad = \mathbf{0}$$

在地面上质心就是重心，否则不一定

# 引力重心




$$\vec{r}_i = \vec{r}_c + \vec{r}_i\,'$$

$$\vec{f} = \sum_i \vec{f}_i = -GM \sum_i \frac{\Delta m_i}{r_i^3}\vec{r}_i$$

一般 $$\neq -GM \frac{\sum_i \Delta m_i}{r_c^3}\vec{r}_c$$

$$r_c >> r_i\,'$$

$$\frac{\vec{r}_i}{r_i^3} = \frac{\vec{r}_c + \vec{r}_i\,'}{(r_c^2 + r_i'^2 + 2\vec{r}_c \cdot \vec{r}_i\,')^{3/2}}$$

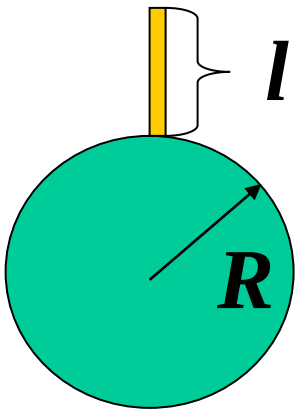
$$= \frac{\vec{r}_c}{r_c^3} + \frac{1}{r_c^3}[\vec{r}_i\,' - \frac{3\vec{r}_c(\vec{r}_c \cdot \vec{r}_i\,')}{r_c^2}]$$

$$\vec{f} = -GM\sum_i \Delta m_i \{\frac{\vec{r}_c}{r_c^3} + \frac{1}{r_c^3}[\vec{r}_i' - \frac{3\vec{r}_c(\vec{r}_c \cdot \vec{r}_i')}{r_c^2}]\}$$

$$\sum_i \Delta m_i \vec{r}_i' = 0$$

$$\vec{f} = -GM\frac{\vec{r}_c}{r_c^3}\sum_i \Delta m_i$$

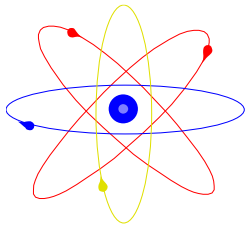
练习




***R>>l***

***R~l***

物体线度远小于到地心的距离时
相当于质量集中于质心



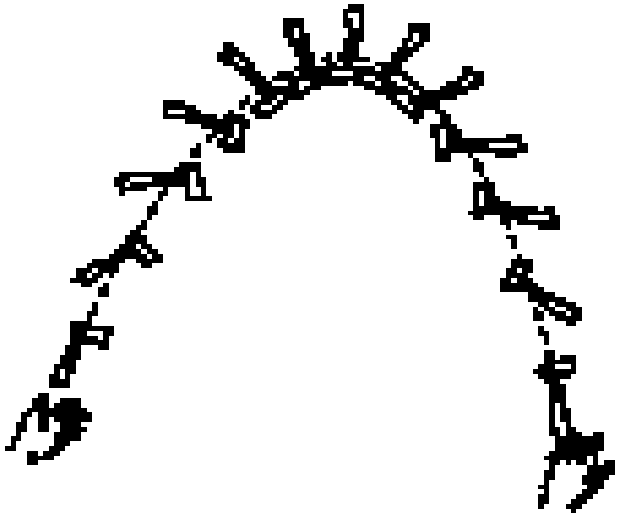
# § 3.6 质心运动定律

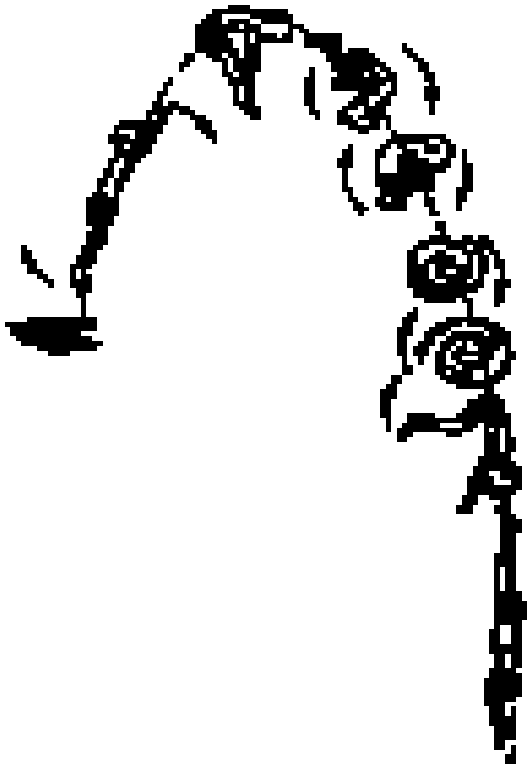
质点系动力学方程 $\vec{F} = \dfrac{d\vec{P}}{dt}$ --------> $\boxed{\vec{F} = M\vec{a}_c}$

$$\vec{P} = \sum_{i=1}^{N} \vec{p}_i = M\vec{v}_c$$

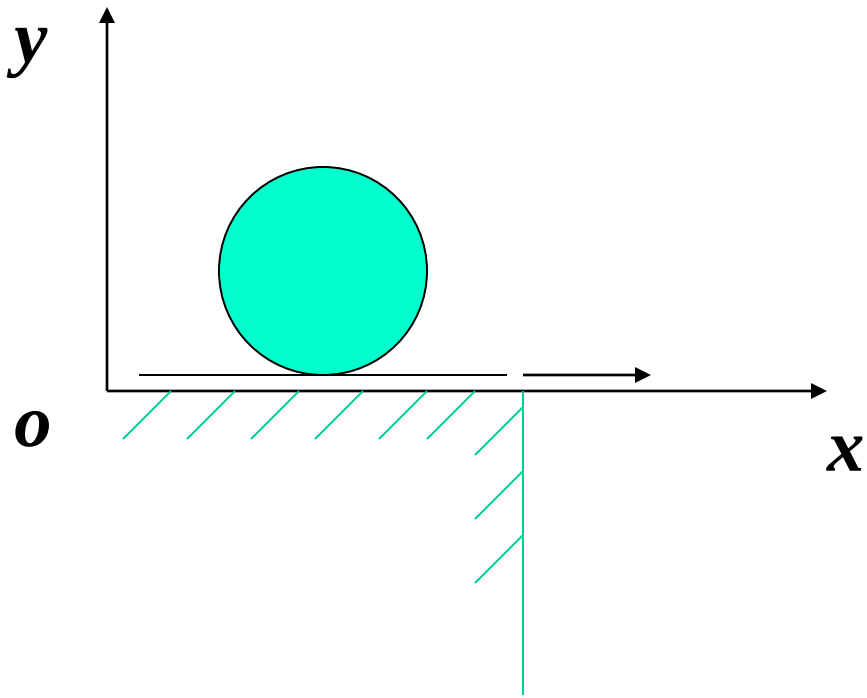
相当于所有质量集中于质心

* 在质心系总惯性力和总外力完全抵消，故动量守恒。





例：水平桌面上拉动纸，纸张上有一均匀球，球的质量$M$，纸被拉动时与球的摩擦力为 $F$，求：$t$ 秒后球相对桌面移动多少距离？




解：

$$\vec{F} = M\vec{a}_c$$

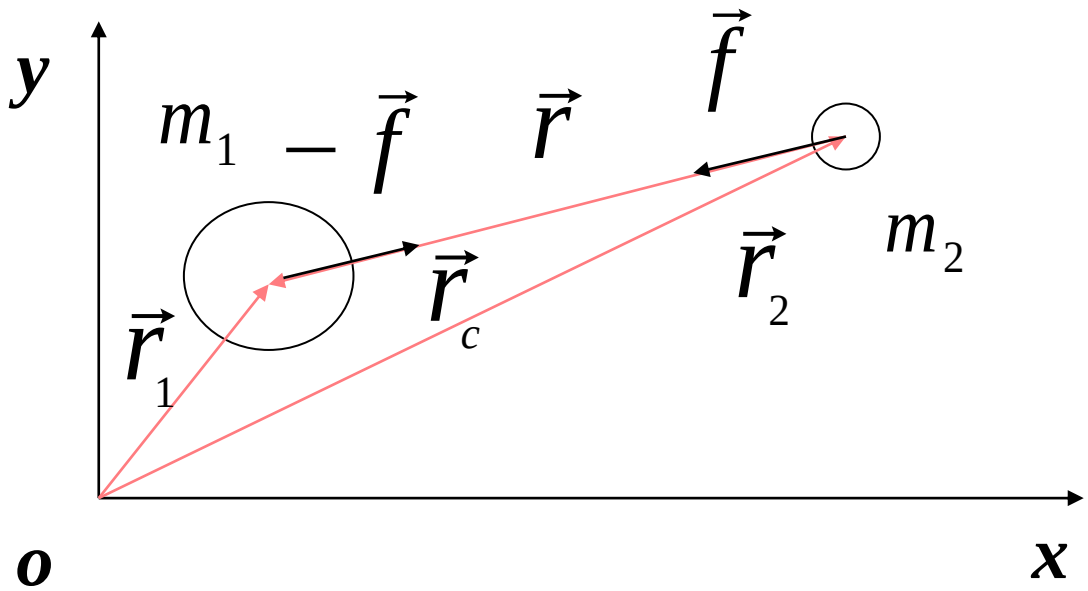
$$a_c = \frac{F}{M} \qquad x_c = \frac{1}{2}\frac{F}{M}t^2$$

$$\frac{1}{2}\frac{F}{M}t^2$$

答：沿拉动纸的方向移动

# §3.7 两体问题

考虑两个物体构成的体系




在惯性参考系

$$-\vec{f} = m_1\vec{a}_1$$

$$\vec{f} = m_2\vec{a}_2$$

$$\vec{a}_c = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2}$$

$$\vec{a}_c = 0$$

## 相对量

$$\vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\vec{v} = \vec{v}_2 - \vec{v}_1$$

$$\vec{a} = \vec{a}_2 - \vec{a}_1$$

$$\vec{a}_1 = -\frac{m_2\vec{a}}{m_1 + m_2}$$

$$\vec{a}_2 = \frac{m_1\vec{a}}{m_1 + m_2}$$

$$-\vec{f} = m_1\vec{a}_1$$

$$\vec{f} = m_2\vec{a}_2$$

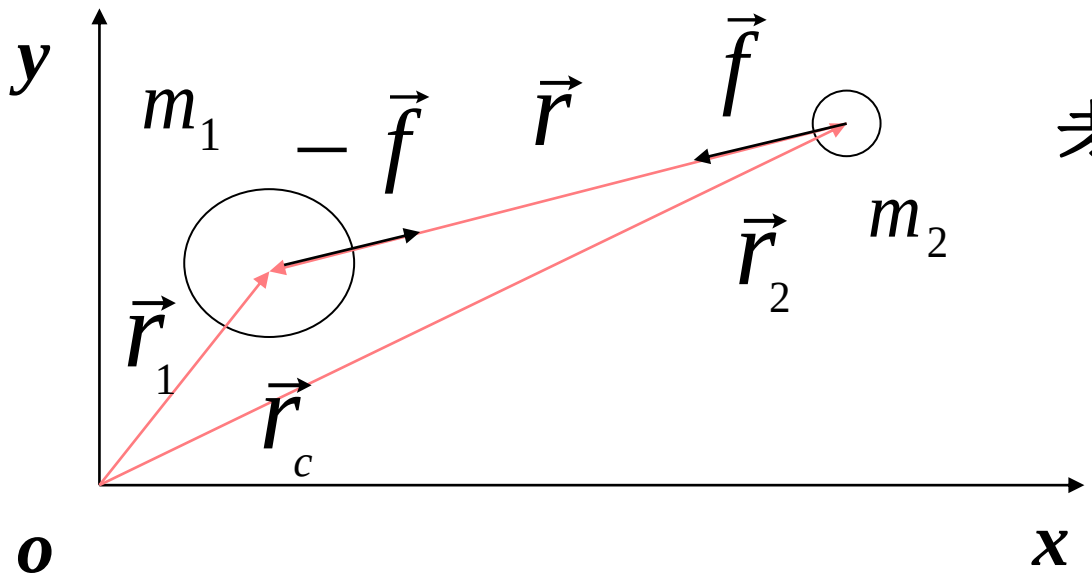
$$\vec{f} = \frac{m_1 m_2}{m_1 + m_2}\vec{a} = \mu\vec{a}$$

$$\boxed{\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}}$$

$\mu$ 称为
折合质量
或者
约化质量

$$\boxed{\vec{f} = \mu\vec{a}}$$

两体问题简化为单体问题。




考虑地绕日运动

$$m_1 >> m_2$$

$$\vec{r}_c = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2} \approx \vec{r}_1$$

$$\vec{a}_1 = -\frac{m_2\vec{a}}{m_1 + m_2} \approx 0$$

$$\vec{f} = \frac{m_1 m_2}{m_1 + m_2}\vec{a} \approx m_2\vec{a}$$

$$\vec{a}_2 = \frac{m_1\vec{a}}{m_1 + m_2} \approx \vec{a}$$

# 氢原子

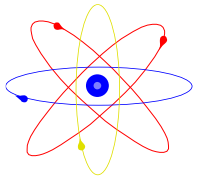
$$-k\frac{e^2}{r^2}=m(\ddot{r}-r\dot{\theta}^2)$$ 近似

严格 $$-k\frac{e^2}{r^2}=\mu(\ddot{r}-r\dot{\theta}^2)$$

$$m\to\mu$$

$\mu$

$$E_n=-\frac{me^4}{2\hbar^2(4\pi\varepsilon_0)^2}\frac{1}{n^2}=-13.6\frac{1}{n^2}\,(eV)\qquad(n=1,2,3,\cdots)$$



# § 3.8 质点的角动量

用叉积定义角动量

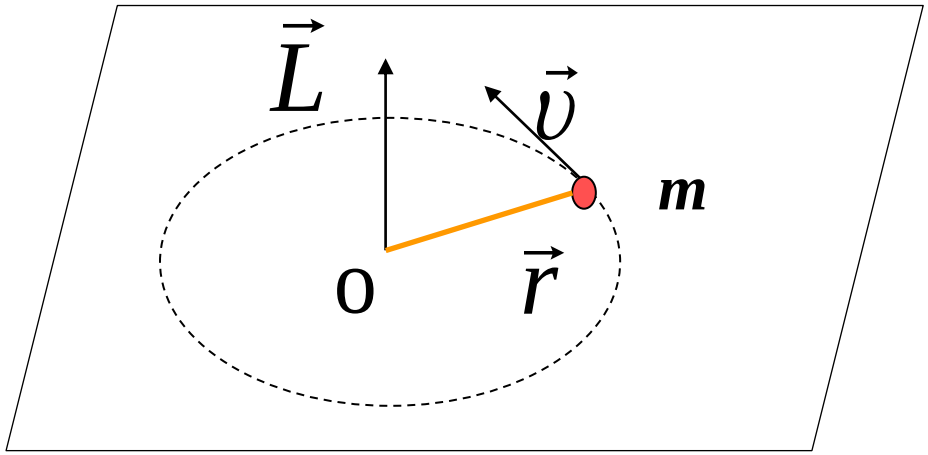
$$\vec{L} = \vec{r} \times \vec{p}$$

相对哪个点？

匀速圆周运动角动量

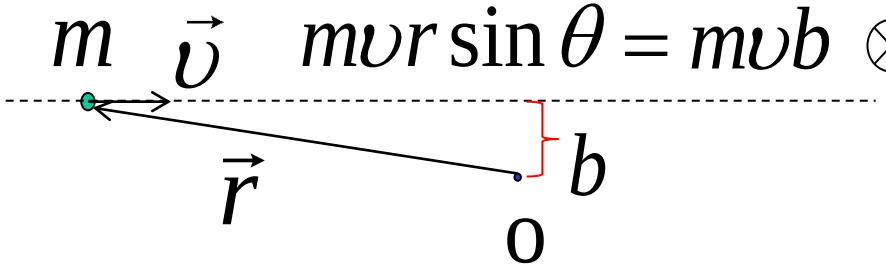
$$L = mr\upsilon = mr^2\omega$$

相对圆心角动量不变




直线运动角动量

$$m\upsilon r\sin\theta = m\upsilon b \otimes$$

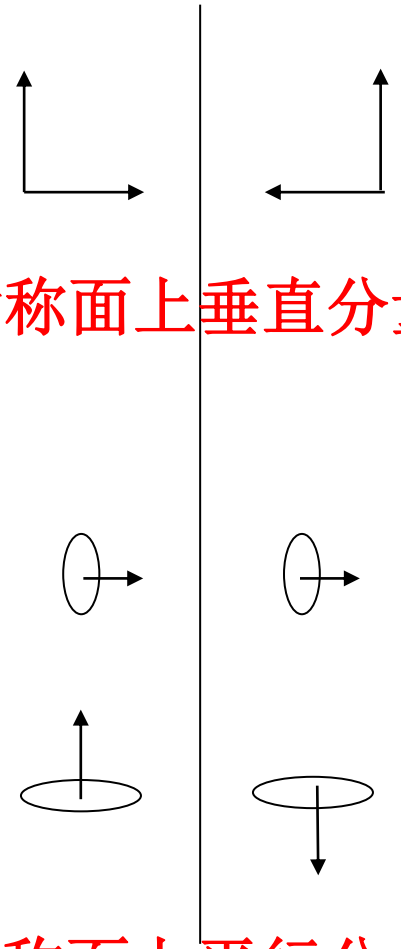


# 镜像对称性

## 对称性: 对某种操作或变换保持不变

### 极矢量

(速度、加速度、电场）



对称面上垂直分量 0

### 镜像变换对称性

### 赝(轴)矢量

(角速度、角动量、磁场）

对称面上平行分量 0

## 角动量定理

$$\boxed{\vec{M} = \dot{\vec{L}} = \frac{d\vec{L}}{dt}} = \frac{d(\vec{r} \times \vec{p})}{dt}$$

$$\dot{\vec{L}} = \overset{0}{\boxed{\dot{\vec{r}} \times \vec{p}}} + \vec{r} \times \overset{\vec{F}}{\boxed{\dot{\vec{p}}}} = \vec{r} \times \vec{F} = \vec{M}$$

$$\boxed{\vec{M} = \dot{\vec{L}}}$$ 相对同一点(惯性系)

$$\boxed{\vec{F} = \dot{\vec{p}}}$$ 等价

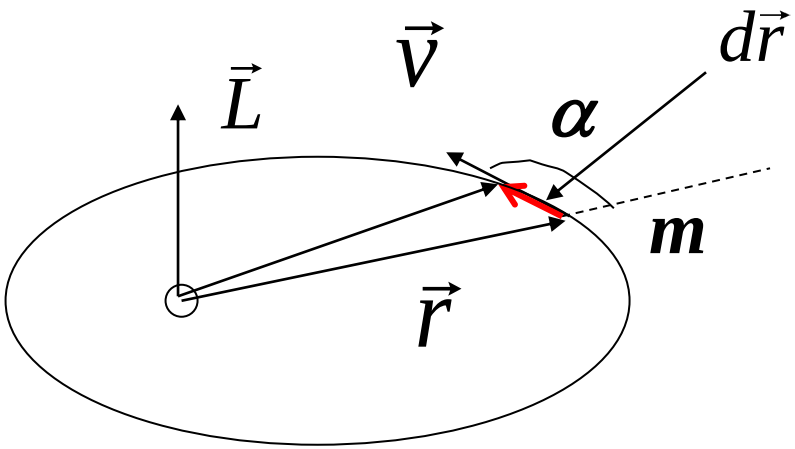
# § 3.9 角动量守恒定律

## 行星运动

行星受力方向与矢径在一条直线（中心力）

$$\vec{M} = \vec{r} \times \vec{F} = 0 \quad \text{----->} \quad \vec{L} = \text{常矢量}$$

故角动量守恒



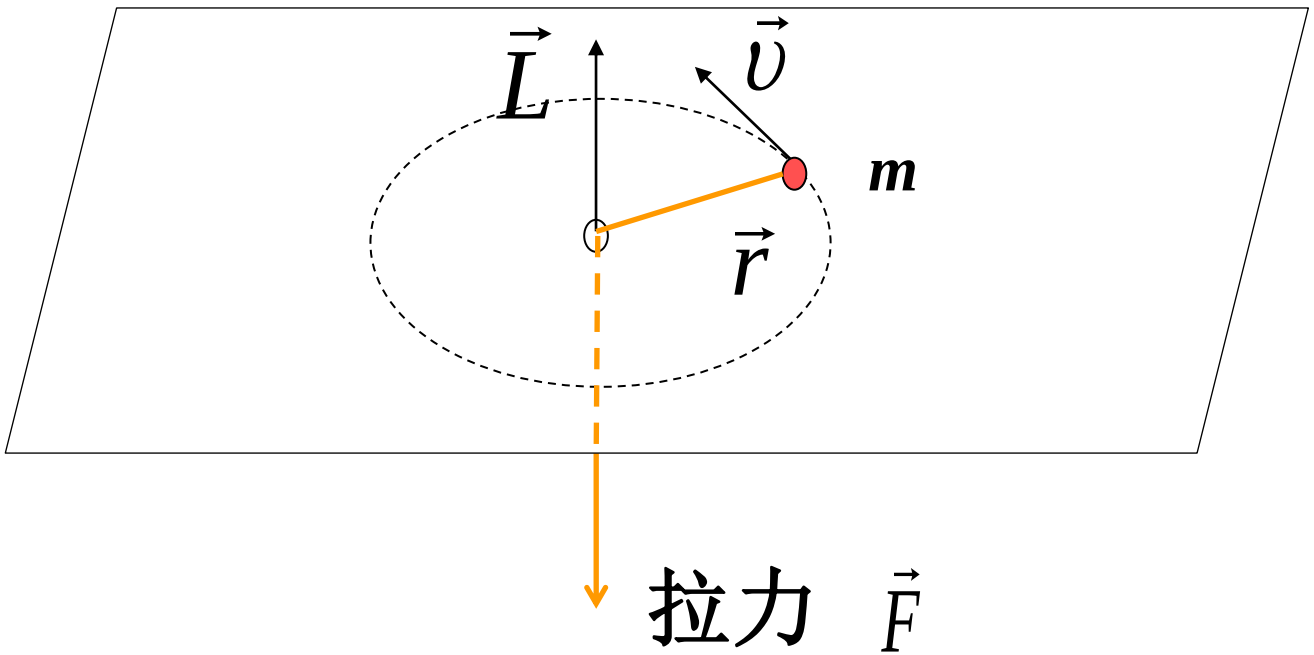

$$\vec{L} = \vec{r} \times \vec{p}$$

$$L = mvr\sin\alpha = m\frac{|d\vec{r}|}{dt} r\sin\alpha$$

$$= 2m\frac{\frac{1}{2}|d\vec{r}|\, r\sin\alpha}{dt} = 2m\frac{dS}{dt}$$

开普勒第二定律

# 角动量守恒演示实验

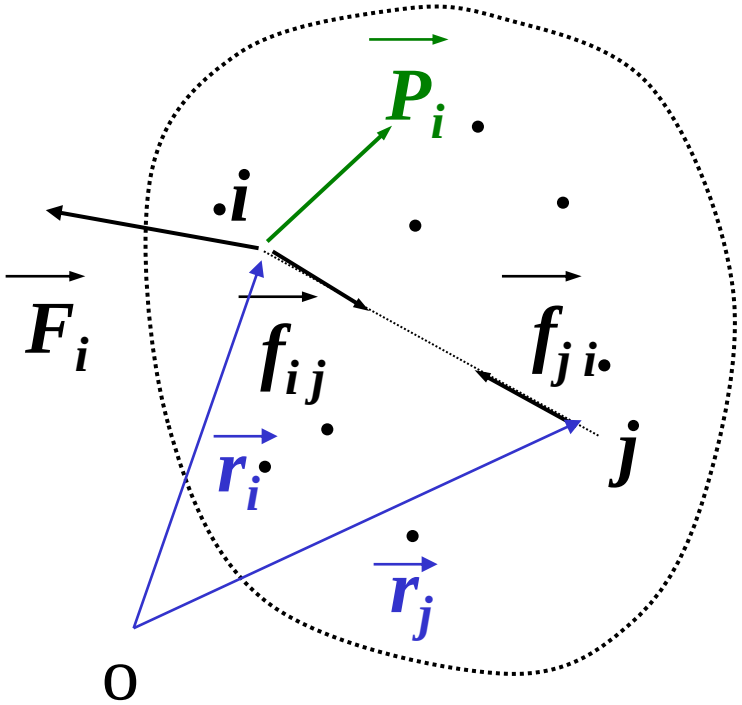



$$\vec{M} = \vec{r} \times \vec{F} = 0$$

$$L = mr\upsilon = mr^2\omega = \text{常量}$$

# 质点系

$$\vec{M}_i = \vec{r}_i \times (\vec{F}_i + \sum_{i \neq j} \vec{f}_{ij}) = \dot{\vec{L}}_i$$




求和 $\sum_i$

$$\vec{r}_i \times \vec{f}_{ij} + \vec{r}_j \times \vec{f}_{ji} =$$

内力矩

$$(\vec{r}_i - \vec{r}_j) \times \vec{f}_{ij} = 0$$

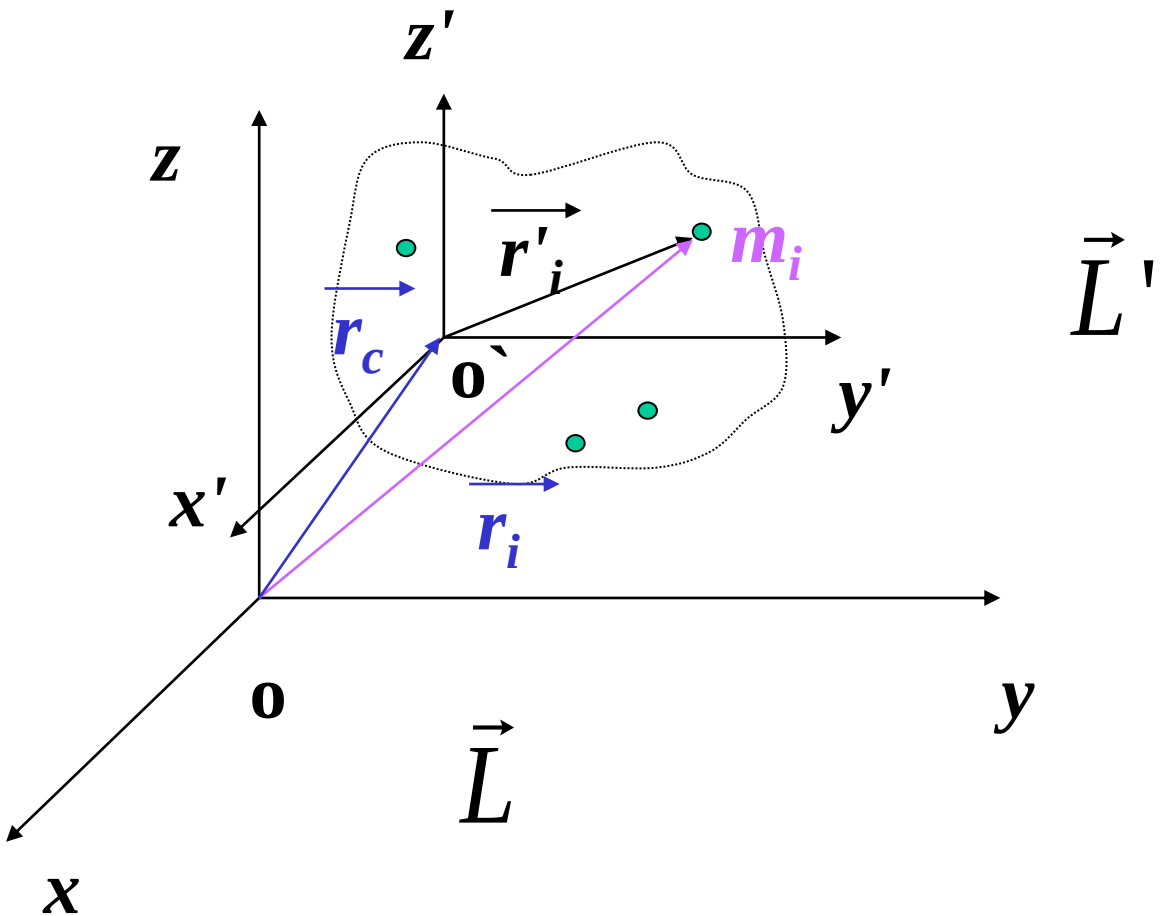
$$\sum_i \sum_{j \neq i} \vec{r}_i \times \vec{f}_{ij} = 0 \qquad \vec{M} = \sum_i \vec{r}_i \times \vec{F}_i = \frac{d}{dt}(\sum_i \vec{L}_i)$$

$$\boxed{\vec{M} = \frac{d\vec{L}}{dt}}$$

质点系的角动量定理

无外力矩，质点系总角动量守恒

# 质心系



$$\vec{L}=\sum_i \vec{r}_i\times\vec{p}_i=\sum_i m_i(\vec{r}_i{}'+\vec{r}_c)\times(\vec{v}_i{}'+\vec{v}_c)$$

$$=\sum_i \vec{r}_i{}'\times\vec{p}_i{}'+\sum_i m_i\vec{r}_i{}'\times\vec{v}_c+\vec{r}_c\times\sum_i \vec{p}_i{}'+\sum_i m_i\vec{r}_c\times\vec{v}_c$$

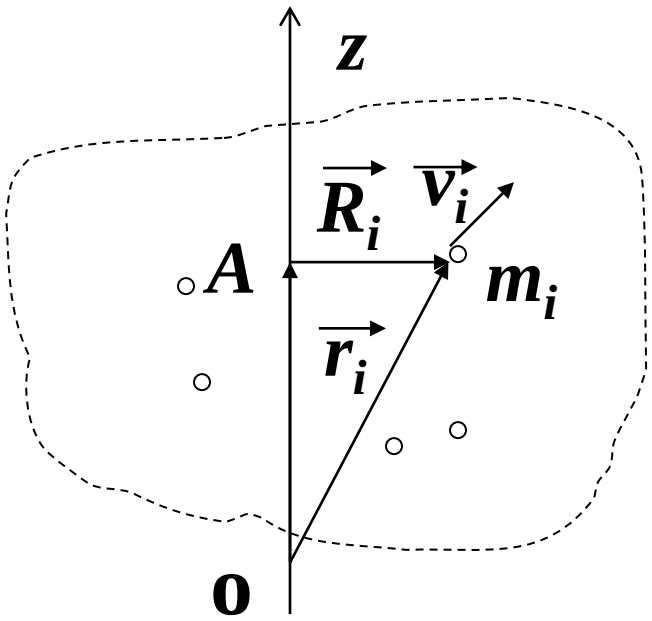
$$\vec{P}=M\vec{v}_c \qquad \boxed{\vec{L}=\vec{L}'+\vec{r}_c\times\vec{P}}$$

$$\dot{\vec{r}}_c\times\vec{P}=0 \qquad \vec{M}=\dot{\vec{L}}'+\vec{r}_c\times\vec{F}$$

质心系 $\boxed{\vec{M}'=\dot{\vec{L}}'}$ 在质心系质点系的角动量定理

# 绕固定轴的力矩和角动量

设固定轴为 $z$ 轴




$$\boxed{M_z = \dot{L}_z}$$

$$L_z = \sum_i (\vec{r}_i \times \vec{p}_i) \cdot \hat{z} = \sum_i (\vec{R}_i \times \vec{p}_{i\perp}) \cdot \hat{z}$$

$$M_z = \sum_i (\vec{R}_i \times \vec{F}_{i\perp}) \cdot \hat{z}$$

$$M_z = 0$$

$$L_z = const.$$

绕固定轴的力矩为 0，
则绕该轴的角动量守恒。

引力使星团压缩，

$$mvr = c$$

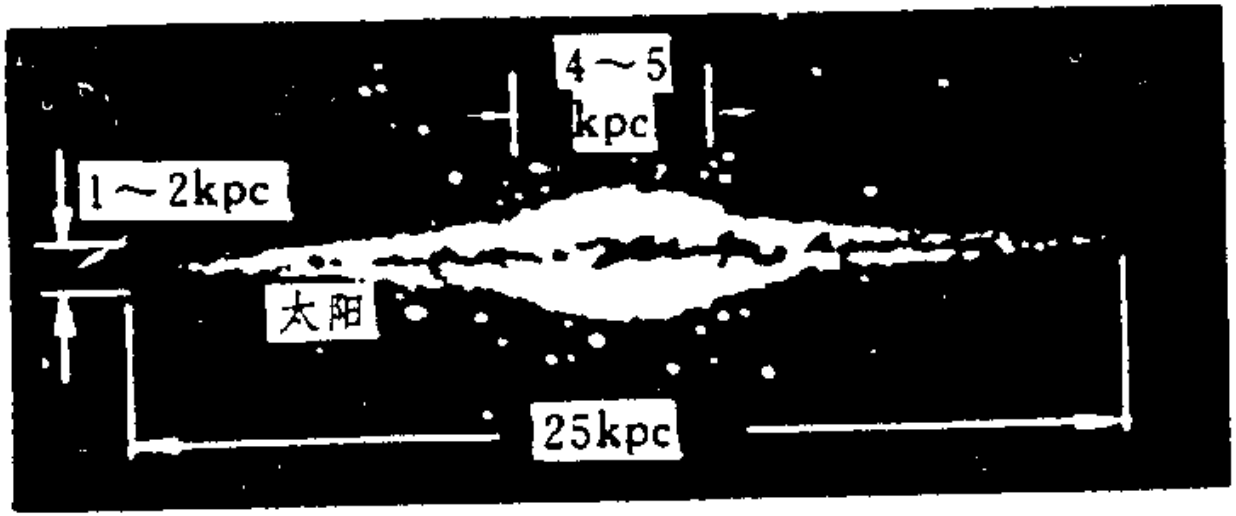
$$v \propto \frac{1}{r}$$

惯性离心力

$$m\frac{v^2}{r} \propto \frac{1}{r^3}$$

离心力与引力达到平衡 ***r*** 就一定了

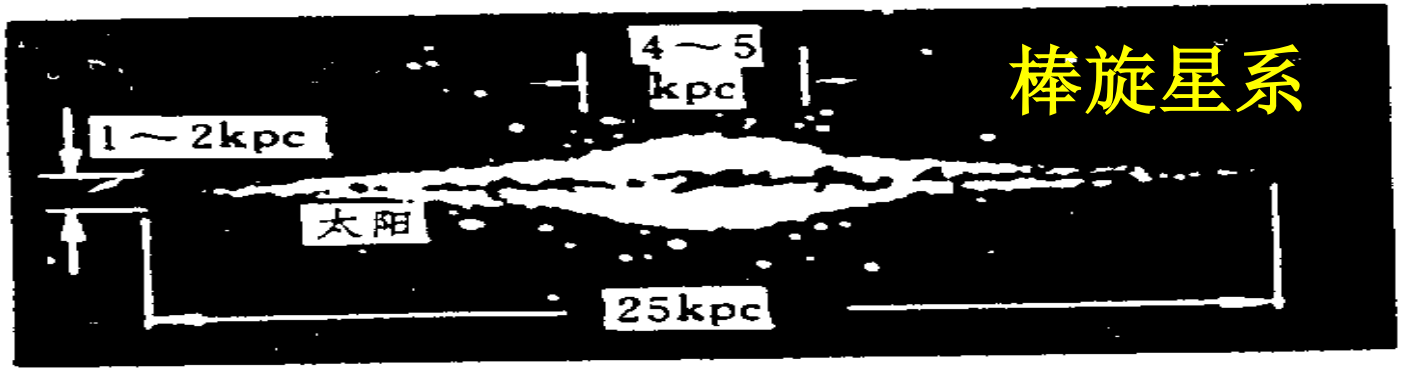
***z*** 轴方向无限制，最终压缩成铁饼状。




1pc~3.26光年

银河系的盘形结构

涡旋星系




棒旋星系

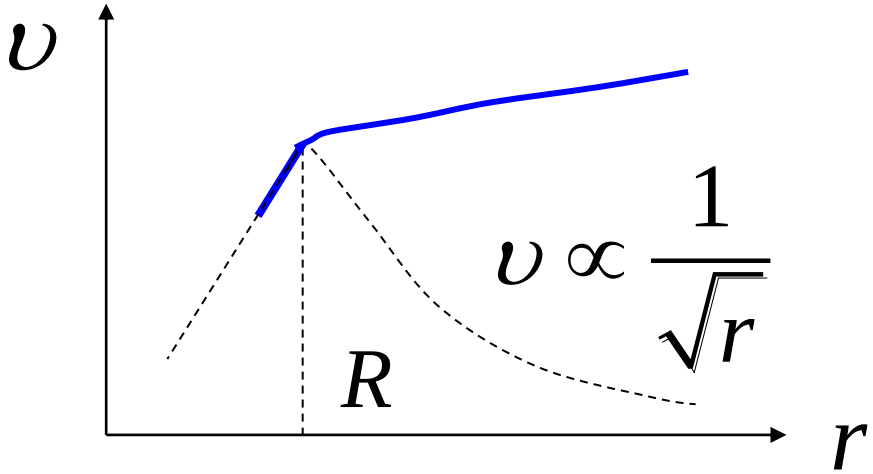
银河系

*1pc(秒差距) = 3.26光年*

速度分布

$$m\frac{\upsilon^2}{r} \propto \frac{1}{r^2}$$

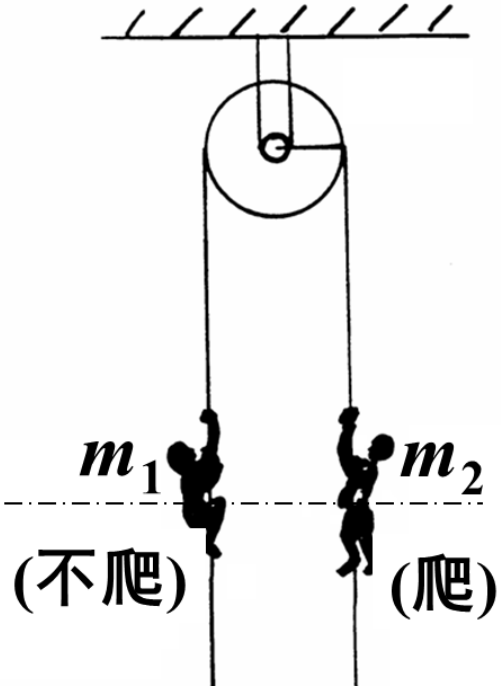
$$\upsilon \propto \frac{1}{\sqrt{r}}$$




观测 *210～240 km/s*

暗物质?

**例** 如图示，两个同样重的小孩，各抓着跨过滑轮的一端。起初都不动，然后右边的小孩用力向上爬绳，左边的小孩只抓住绳但不爬动。忽略滑轮、绳的质量和轴的摩擦。设他们的起始高度相同，问哪个小孩先到达滑轮？




解答：小孩重力相等，相对轴心力矩大小相等方向相反，那么整个系统合外力矩为零，**角动量守恒**。

**相对于轴心，**

$$mv_1r = mv_2r \qquad v_1 = v_2$$

两个小孩同时到达滑轮。

**例** 光滑平面上，质量均为 $M$ 的两小球由一长为 $l$ 的轻杆相连. 另一个质量为 $m$ 的小球与某一 $M$ 发生完全非弹性碰撞. 所有小球的大小可以忽略.

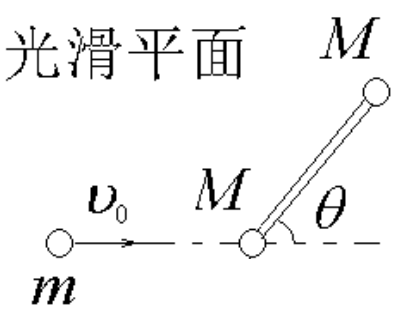
试问:

1) 若以碰撞点为原点，则相对原点碰撞前后系统角动量是否守恒？碰撞瞬间杆的另一端 $M$（没有与$m$直接碰撞）的速度方向？

2) 碰撞后轻杆系统绕其质心转动的角速度 .

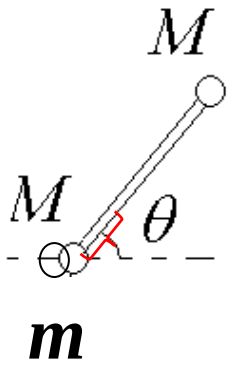
$$m = M, \quad \theta = 45^o$$

**解:** 1) 角动量守恒, 另一端 $M$ 沿杆方向




2) 杆系统角动量相对原点为零

质心速度 $$\vec{V}_c = \frac{1}{3}\vec{\upsilon}_0$$




质心 $y$ 坐标

$$y_c = \frac{l\sin\theta}{3}$$

$m = M, \quad \theta = 45^o$

碰后系统角动量

$$\boxed{\vec{L} = \vec{L}' + \vec{r}_c \times \vec{P}}$$

$$-(M+m)\frac{1}{3}l\omega\frac{1}{3}l - M\frac{2}{3}l\omega\frac{2}{3}l + (2M+m)\frac{1}{3}\upsilon_0\frac{1}{3}l\sin\theta = 0$$

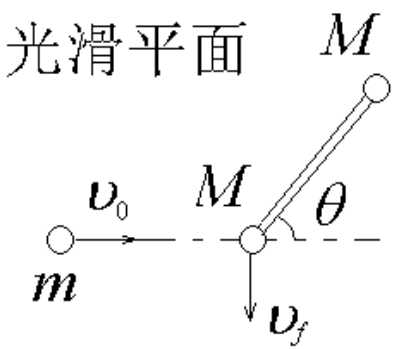
$$\omega = \frac{\sqrt{2}}{4}\frac{\upsilon_0}{l}$$

**例** 光滑平面上，质量均为 $M$ 的两小球由一长为 $l$ 的轻杆相连. 另一个质量为 $m$ 的小球与某一 $M$ 发生完全弹性碰撞，碰后 $m$ 沿垂直于原速度方向运动，如图所示. 所有小球的大小可以忽略.

试问: 碰撞后$m$的速度 和轻杆系统绕其质心转动的角速度 .

$$m = M, \quad \theta = 45^o$$

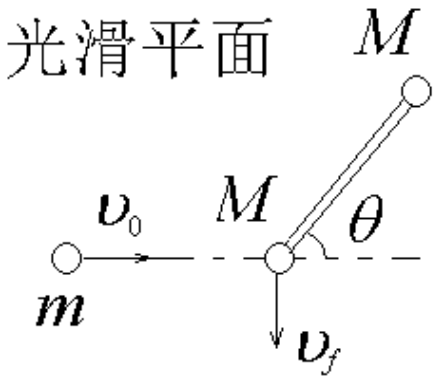
**解:** 角动量守恒




杆系统角动量相对原点为零

碰后杆质心速度

$$\vec{V}_c = \vec{V}_{//} + \frac{1}{2}\vec{V}_{\perp}$$




## 杆角动量

$$\boxed{\vec{L} = \vec{L}' + \vec{r}_c \times \vec{P}}$$

$$-2M\frac{1}{2}l\omega\frac{1}{2}l + 2M\frac{1}{2}V_{\perp}\frac{1}{2}l = 0$$

$$\omega = \frac{V_{\perp}}{l}$$

$$m = M, \quad \theta = 45^{o}$$

$$m\upsilon_0\cos\theta = 2MV_{//} - m\upsilon_f\sin\theta$$

$$m\upsilon_0\sin\theta = MV_{\perp} + m\upsilon_f\cos\theta$$

$$\frac{1}{2}m\upsilon_0^2 = \frac{1}{2}MV_{\perp}^2 + MV_{//}^2 + \frac{1}{2}m\upsilon_f^2$$

$$V_{\perp} = \frac{3\sqrt{2}\pm 2}{7}\upsilon_0 \qquad V_{//} = \frac{2\sqrt{2}\mp 1}{7}\upsilon_0 \qquad \upsilon_f = \frac{1\mp 2\sqrt{2}}{7}\upsilon_0$$

$$\omega = \frac{3\sqrt{2}-2}{7}\frac{\upsilon_0}{l}$$

第三章完

本章编者

安宇